and the division of β_t in Eqn 28 normalizes the gradient norm at each position t as we will later see in Eqn 31 and also clarified in Section 4.1 (and Appendix D.2).

Note that $\beta_t \cdot D_{\text{KL}}\Big(\pi^*(\cdot|\boldsymbol{x}, \boldsymbol{y}_{<t}) \Big\| \pi_{\text{aligned}}(\cdot|\boldsymbol{x}, \boldsymbol{y}_{<t})\Big) \geq 0$ is a non-negative constant, we have:

$$\mathbb{P}_\theta\left(T([\boldsymbol{x}, \boldsymbol{y}_{<t}], y_t)\right) \geq \sigma\left(\beta_t \cdot \log \frac{\pi_\theta(y_t|\boldsymbol{x}, \boldsymbol{y}_{<t})}{\pi_{\text{aligned}}(y_t|\boldsymbol{x}, \boldsymbol{y}_{<t})}\right) \tag{30}$$

So, the objective in Eqn 28 can be replaced with a lower bound surrogate objective L_θ:

$$\mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left\{\sum_{t\geq 1}\frac{1}{\beta_t}\log \mathbb{P}_\theta\left(T([\boldsymbol{x}, \boldsymbol{y}_{<t}], y_t)\right)\right\} \geq L_\theta \tag{31}$$

where

$$L_\theta := \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left\{\sum_{t\geq 1}\frac{1}{\beta_t}\cdot \log\ \sigma\left(\beta_t \cdot \log \frac{\pi_\theta(y_t|\boldsymbol{x}, \boldsymbol{y}_{<t})}{\pi_{\text{aligned}}(y_t|\boldsymbol{x}, \boldsymbol{y}_{<t})}\right)\right\}.$$

Eqn 3 is then equivalent to $\min_\theta -L_\theta \iff \max_\theta L_\theta$. So, optimizing the objective Eqn 3 is essentially to maximize the lower-bound of the reinforcement learning objective in Eqn 28.